

10 Questions to Fall in Love With A Chatbot: An Experimental Study on Interpersonal Closeness With Large Language Models (LLMs)

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Abstract

Large language models (LLMs), like ChatGPT, are able to compute affectionately nuanced text, which can be deployed in online dating or power artificial companions. This study explores whether people can experience intimate interest and interpersonal closeness toward a potential human or AI partner, based on such text. In a mock matchmaking scenario, 307 participants rated 10 text responses to the Interpersonal Closeness Generation Task from the dating profile of their potential partner, unaware that all were LLM-generated. Surprisingly, the source label (human or AI) had no significant impact on intimate interest and interpersonal closeness. Instead, the perceived quality and human-likeness of responses shaped reactions. These findings advance research in human-machine communication, with social and ethical implications for the future of our AI-mediated intimacy.

Keywords: large language models, LLM, online dating, intimacy, ChatGPT, interpersonal closeness

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Introduction

Recent progress in large language models (LLMs) is enabling the advent of systems with natural language capabilities that rival those of humans. Communication being at the foundation of our social relationships, these technological advances are reshaping both, how and with whom humans can establish intimate connections (Dubé & Anctil, 2020). These systems can serve several functions, such as practicing sociosexual interactions (e.g., flirting), enhancing communications (e.g., crafting more compelling texts), and powering artificial companions (cf., Nomi.AI, 2024; Replika, 2024). In that regard, early surveys show that some people already want to use AI to craft love letters and struggle to distinguish between those written by AIs and those written by humans (McAfee, 2023). Moreover, the fifth most downloaded smartphone application in 2024 was RIZZ Text, an LLM-powered “dating assistant” that is built to “generate personalized responses that are sure to impress your crush” (Predin, 2024; Wega Labs, 2026). More recent evidence also highlights that 26% of US singles are using AI to enhance their dating life (a 333% year-to-year increase), while 16% have engaged with AI as a romantic companion (Match.com, 2025).

What may facilitate online dating success or intimate companionship access raises several questions about the effects that AI-enhanced communications and human-machine relationships may have on perceived transparent, genuine interhuman communication, as well as the possibility and process through which people may form romantic and sexual relationships with machines. These issues are especially pressing considering that meeting partners online is now the most prevalent way to find potential partners and build intimate connections (Castro & Barrada, 2020). Given the potential to disrupt norms and behaviors around online dating communication, and the rapid spread of statistically optimized LLM-powered artificial companions, this study investigates whether the source label of a message (i.e., human or artificial) and its perceived qualities affect first impressions of a potential intimate partner, as measured through interpersonal closeness and intimate interest.

To do so, this study employed a mock AI-matchmaking scenario combined with an adapted version of the Interpersonal Closeness Generation Task (ICGT, commonly referred to as the *36 questions to fall in love*, Aron et al., 1997). In this version, participants first provided their own answers to 10 ICGT questions. They were then presented with answers to the same set of questions from their allegedly perfect match (i.e., a human or AI partner presented in a dating profile format), creating the impression of a reciprocal interaction as typically found on dating or matchmaking websites. Importantly, regardless of whether they were presented as coming from a human or AI, all answers from this “perfect match” were generated by the freely available text-generative chatbot, ChatGPT 3.5., and did not differ between the two conditions (OpenAI, 2023; see Figure 2 for an answer example). In doing so, this study contributes to a deeper understanding of whether people could develop feelings of interpersonal closeness and intimate interest toward LLM-generated text if it was indistinguishable in text-based communications (i.e., the human partner condition, which is LLM-generated). It also examines whether the source label and perceived qualities of communication influence the capacity to evoke such feelings in an asynchronous, one-shot reading task taking place within a matchmaking scenario.

Literature Review

Theoretical Frameworks

Communication is shaped and co-constructed by interpretations of the situation and communication partners, which is itself influenced by the perceived or interpreted ontology of the participants (Edwards & Etzrodt, 2025). While interhuman communication involves partners belonging to the same ontological category, human-machine communication (HMC) involves meaning-making and world-construction processes between entities that stem from different modes of being (Dautenhahn, 2004; Edwards & Etzrodt, 2025). Early HMC paradigms, such as Media Equation Theory and Computers as Social Actors, approached human-machine communication as *cue-response* processes, whereby machine behavior causes automatic (or mindless) social response pattern from human (Edwards & Etzrodt, 2025; Nass & Moon, 2000; Reeves & Nass, 1996). However, these paradigms often fail to account for how people make sense of machines, their behavior, and their contextual engagement, as well as how this sense-making is constitutive of the emergent human-machine relationship (Edwards & Etzrodt, 2025). In response, more recent paradigms such as Media Evocation reinterpret machines as *evocative objects*: entities that elicit reactions through their symbolic meaning and contextual engagement (van der Goot & Etzrodt, 2023).

Building on this premise, the Act-Mean-Relate (AMR) paradigm specifies how such evoked responses are translated into relational perceptions: a recursive loop between agents that considers what communicative partners, including machines, do (act), how their actions are interpreted (meaning), and how they occur within and are constitutive of the social relation (Edwards & Etzrodt, 2025). According to this paradigm, understanding LLM-enhanced online dating and HMC within intimate relationship building requires us to not only consider the agents' actions (e.g., what partners say or write), but their interpretation (e.g., the meaning attributed to the agent and its actions), the contextual engagement (e.g., situational and sociocultural factors), and the type of relationship in which communication unfolds and that emerges from it (e.g., platonic or romantic).

When considering intimate and affectively charged communication with artificial entities, the AMR framework resonates with existing theoretical models, including the Human-Erobot Interaction and Co-Evolution Model (HEICEM; Dubé & Anctil, 2020) and the Sexual Interaction Illusion Model (SIIM; Szczuka et al., 2019). The HEICEM posits that humans and artificial agents co-construct relationships through interactions, such as experiences of positive or negative communication, along with the meaning attributed to these interactions and the communicative agents, and the transformation of the ecological niche in which they are embedded and of which they are constituents (Dubé & Anctil, 2020). This ranges from its microsystems (e.g., proximal friend, partner, and computer networks) and exosystems (e.g., political, education, legal, and health institutions) to its macrosystems (e.g., sociocultural norms and values; Dubé & Anctil, 2020).

Complimentarily, the SIIM emphasizes that social responses to artificial partners depend on both automatic and reflective processing (Szczuka et al., 2019). For example, sexual arousal and intimate interest toward artificial partners emerge from the interplay of fast, largely automatic responses to sexual cues and more reflective, socioculturally shaped

evaluations. Crucially, these processes are influenced by source perception, as the perceived artificial nature of a partner can shape both immediate reactions and more deliberative judgments, such as the moral acceptance of human machine intimacy grounded in learned social scripts (Szczuka et al., 2019). Thus, willingness to engage intimately with an artificial partner depends on the quality of the affordance-based cues provided by the machine, knowledge about the agent's nature, the meaning attributed to this information, and the time available to process it (Szczuka et al., 2019).

Together, these frameworks indicate that understanding people's reactions and behavior within technology-enhanced and (intimate) HMC requires considering multiple factors. These include the ontological category of the communicative agent and the qualities of the communication itself. In addition, meaning-making is shaped by the situational and sociocultural context in which communication is embedded, as well as by how these elements are interpreted and evaluated (Dubé & Anctil, 2020; Edwards & Etzrodt, 2025; Szczuka et al., 2019). To date, there is little systematic empirical work examining intimacy relevant impressions in text-based HMC under conditions where message content is held constant and only the labeled source varies. From a Media Evocation and AMR perspective, this leaves open whether early feelings of closeness and intimate interest are primarily shaped by categorical source labels or by the meanings attributed to communicative acts, including perceived human-likeness and authenticity cues, a distinction that becomes increasingly consequential as LLMs are more frequently embedded in dating and matchmaking contexts.

Empirical Insights

The scarce research on interpersonal closeness and intimate interests toward humans compared to machines (e.g., robots, chatbots, and virtual companions) suggests that, when someone knows the disclosed nature of an agent, they tend to prefer humans over artificial partners, especially in romantic and/or intimate contexts (Dubé et al., 2022; Szczuka, 2022; Szczuka & Szymczyk, 2024). For instance, an empirical study comparing the flirtatiousness of voice assistants to identical content delivered as a voice message from a human showed that the human voice was perceived as more sexually attractive and flirtatious (Szczuka, 2022). The author argued that flirtatiousness signals a human drive and interest toward others that can only be (perceived as) genuinely felt by another human. Humans may also prefer other humans for reasons such as reproductive purposes, societal norms discouraging nonhuman intimacy, and the current shortcomings of technologies that prevent more complex intimate interactions (e.g., Banks & Van Ouytsel, 2020; Szczuka et al., 2019). Moreover, human relationships with artificial partners are often portrayed and discussed as a "last resort" for individuals who struggle with social interactions with other humans, including but not limited to those with intellectual and physical disabilities or neurodiverse individuals (Döring & Poeschl, 2019; Turkle, 2011). This may trigger responses based on social desirability and the reflex of separating oneself from such stereotype, manifesting in negative reactions toward intimacy with technology (Mühl & Szczuka, 2024). First evidence of such stigma was empirically demonstrated by Dubé and colleagues (2023) who found that the more humanlike an erotic technology is (including virtual partners), the more perceived stigma its use elicits.

Contrary and despite such stigma, some people seek artificial partners precisely for their advantages and the unique relationships that they may provide, fully aware of their companion's nature (Lievesley et al., 2023). Artificial agents can, for instance, facilitate continuous access to judgment-free communication and intimacy (Dubé & Anctil, 2020). This can include communication about various romantic and sexual desires, which may deviate from traditional or conservative sociosexual scripts and norms, and could otherwise be kept secret to avoid societal backlash (Dubé & Anctil, 2020). This notion is further supported by the finding that romantic fantasy may be central in these relationships, as it enables judgment-free exploration of desires and supports strong personalization where users can actively co-create an idealized, emotionally responsive partner (Ebner & Szczuka, 2025). Pentina and colleagues (2023) further found that human-likeness and authenticity, here meaning perceived agency, autonomy, and uniqueness, play a key role in accepting an AI-driven interaction partner.

Relatedly, other evidence suggests that the perceived or labeled nature of an agent may have little to no effect on people's responses. It is rather variables pertaining to the communication qualities that seem to take primacy (A. Ho et al., 2018). For example, A. Ho and colleagues found no effect of perceived partner identity (i.e., human or chatbot) on emotional, relational, and psychological variables. Participants seemed to benefit just as much from self-disclosing to a supposed chatbot as to a supposed human, and conversational processes seem to have unfolded similarly regardless of partner identity (A. Ho et al., 2018). It is rather the qualities of the disclosure (i.e., emotional or factual) that significantly affected variables like mood, perceived warmth, conversational enjoyment, partner liking, perceived understanding, self-affirmation, disclosure intimacy, and cognitive reappraisal (A. Ho et al., 2018). Taken together, existing empirical findings present a mixed picture. While some studies point to a primacy of communication qualities over agent identity, others highlight the continued relevance of perceived ontology in intimate contexts. Against this backdrop, research has yet to systematically examine how source-labeled ontology and communication qualities jointly shape human-machine communication in intimate settings, or how individual and sociocultural factors contribute to the emergence of interpersonal closeness and intimate interest.

Influencing Variables

Previous research suggests that gender and sociocultural contexts may shape reactions toward artificial partners. Indeed, one of the most robust findings stemming from the burgeoning body of research on sexualized interaction technology is that men are more interested in intimacy with machines than women (e.g., sex [with] robots; Bame, 2017; Brandon et al., 2022; Brandon & Planke, 2021; Dubé et al., 2023, 2024; Hanson, 2013; Nguyen, 2020; Nordmo et al., 2020; Oleksy & Wnuk, 2021; Scheutz & Arnold, 2016). This may stem from a combination of gynoid marketing targeting heterosexual men, media representations mostly depicting men-gynoid intimate relationships, societies devaluing women's sexuality while valuing men's, as well as discrepancies in sexual drive, risk-taking, and sensation-seeking (cf., Dubé et al., 2022; Peixoto, 2023). Regardless of the reason, this may translate into a propensity for men to experience greater feelings of interpersonal

closeness and interest in intimate interactions with both human and artificial partners than women in this study.

In parallel, grounded in location and cultural values/dimensions, nationality-based differences may exist between North American and European countries, like Germany and Canada, influencing people's reaction toward human-machine intimacy. In that regard, and acknowledging that both are multicultural societies, Hofstede's cultural dimensions (Hofstede & Bond, 1984) indicate that, while Germany and Canada exhibit some similarities, they may also present differences in indulgence-restraint (i.e., the extent to which they attempt to control their desires and impulses) and uncertainty avoidance (i.e., their discomfort with ambiguous, unknown, or unfamiliar situations). These differences may influence Germans' and Canadians' respective reactions to artificial partners, particularly in matters of intimacy and sexuality. In practice, this may manifest by Canadians being generally more accepting of new ideas or products and more willing to try something new, while Germans tend to enjoy detailed and well-thought-out concepts. Applied to the context of potential intimate communication with artificial partners, these differences between Germans and Canadians on Hofstede's cultural dimensions would likely translate into Canadians being more open to forming interpersonal closeness and intimate interest in machines than Germans.

While the variables described above were included to test interaction effects, the study further considered several individual difference variables as potential covariates. Drawing on the illustrated prior research above, these include loneliness, attitudes toward artificial intelligence, liking of the response content, and participant age. Rather than including all variables by default, covariates were selected based on preliminary analyses assessing their statistical relevance and overlap, in order to maintain model parsimony and avoid multicollinearity.

The Current Study

The present study aimed to experimentally examine whether individuals develop interpersonal closeness and intimate interest toward a potential dating partner based on LLM-generated self-disclosure, depending on whether the partner is labeled as human or AI (**Objective 1**). In addition, the study examined whether these effects vary as a function of participant gender and nationality (**Objective 2**).

Based on prior research on HMC and intimate interaction technologies, we hypothesized that participants would report greater interpersonal closeness and interest in intimate interactions when they believed the dating profile represented a human rather than an AI partner (**Hypothesis 1**). We further hypothesized that men, compared to women, would report greater interpersonal closeness and interest in intimate interactions across both partner conditions (**Hypothesis 2**). Finally, we hypothesized that Canadian participants would report greater interpersonal closeness and interest in intimate interactions with an AI partner than German participants, while no nationality differences were expected for the human partner condition (**Hypothesis 3**).

In addition to these confirmatory hypotheses, the study included exploratory analyses to examine whether individual difference variables, such as loneliness, attitudes toward

artificial intelligence, liking of the response content, and age, were associated with interpersonal closeness and intimate interest.

Methods

This study was approved by the ethics Committee of the Faculty of Computer Science, University Duisburg-Essen, Duisburg, Germany (ethics certification number: 2304SPSJ5267). Informed consent was obtained electronically from each participant prior to study completion. Preregistration, data, and material are available on the Open Science Framework (https://osf.io/jqv9k/overview?view_only=1d57a04659e04a4797873ce8606a670b).

Participants

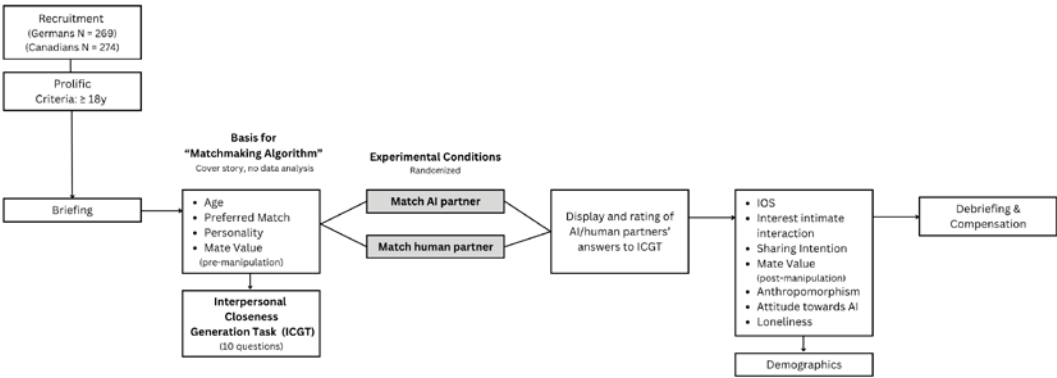
A power analysis was conducted using G*Power to determine the required sample size. The objective was to achieve a power of .95 to detect an effect size of $f^2(V) = .04$ (small to medium effect) with an alpha level of $\alpha = .05$. The recommended sample size of the utilized multivariate analysis of variance and covariance (MANCOVA) was found to be 347. A convenience sample of 274 Canadians and 269 Germans was initially recruited on Prolific in June 2023. Multiple attention and manipulation checks were performed to ensure participants paid attention during the study. The answers of 101 German and 91 Canadian participants had to be removed from the analysis because they failed the manipulation check, incorrectly identifying the ontological class of their match. Because the source label of the profile is the key to the study, this hard cutoff was needed to ensure that people were aware of their assigned condition. The final sample consists of 307 participants (AI label: $n = 173$; human label: $n = 134$), of whom 142 were Germans (46.2%) and 165 were Canadian (53.8%). In terms of gender, the sample comprised 163 males and 144 females, with ages ranging from 19 to 79 ($M = 34.03$, $SD = 10.76$).

Procedure

To test the hypotheses, this study used a 2 (source label: human or artificial partner) $\times 2$ (gender: male or female) $\times 2$ (nationality: German or Canadian) between-subject design. This study was framed as an “online dating/matchmaking scenario.” Participants were told they would engage with a novel proprietary matchmaking algorithm designed to pair them with a compatible potential partner. The central task revolved around 10 self-disclosure questions adapted from the ICGT (Aron et al., 1997).

Participants were first asked to provide information about their age, personality, and preferred match (i.e., gender and sexuality of people they are interested in). They were then instructed to write their own answers to 10 ICGT questions, simulating the self-disclosure process. Once they were done, participants were informed that they had been matched with either an AI or a human partner (source-label condition). They were then shown the ICGT responses from their assigned match and asked to rate the appeal of each response. No live interaction occurred. Participants were presented with prewritten responses from what seemed like the existing profile of their assigned match, comparable to those found

FIGURE 1
Experimental Design



on dating applications (see Figure 1 for an experimental design overview and Figure 2 for an example of response).

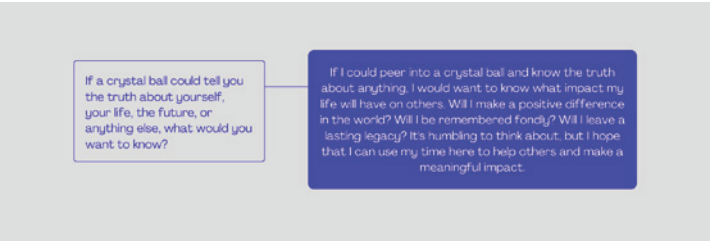
Following this, participants answered questions about their level of interpersonal closeness and interest in engaging in further intimate interactions with the matched partner. Participants also answered additional measures (e.g., loneliness) and demographics (see Measures section for details). At the end of the study, participants received a debriefing and their compensation.

Measures

Interpersonal Closeness

The Inclusion of Other in the Self Scale (IOS; Aron et al., 1992; cross translation was used for the German version) was used to assess participant-match interpersonal closeness. This is a single-item measure asking participants to select between seven pairs of circles varying in overlap, ranging from barely touching to almost fully overlapping. One circle was labeled “me” and the other one was labeled “partner.” Participants were asked to select the pair that best represented their perceived connection with the match.

FIGURE 2
Example for one of the answers to the ICGT provided by the LLM ChatGPT 3.5



Intimate Interest

The Erotic Engagement Questionnaire (EEQ; Dubé et al., 2022; cross translation was used for the German version) was used to measure the intimate interest toward the assigned match. This is a 4-item measuring interest in love, sex, friendship, and intimate relationships with a partner (e.g., “Would you engage in an intimate relationship with . . .?”) on a 5-point Likert scale (1 = not at all, 5 = definitely). This measure showed good internal consistency ($\alpha = .86$).

Loneliness

The short version of the UCLA Loneliness Scale (ULS-8; Hays & DiMatteo, 1987; cross translation was used for the German version) was used to assess feelings of loneliness among participants. This is an 8-item scale (e.g., “I lack companionship” or “I feel left out”) employing a 5-point Likert scale (1 = never, 5 = always). This measure showed good internal consistency ($\alpha = .90$).

Attitude Toward Artificial Intelligence

The General Attitudes toward Artificial Intelligence Scale (GAAIS; Schepman & Rodway, 2023; German version was obtained from Sindermann et al., 2021) was used to assess perceptions of AI. This is a 20-item scale (12 positive items, such as “Artificial Intelligence is exciting” and 8 negative items, such as “I think Artificial Intelligence is dangerous”) employing a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). This measure showed good internal consistency ($\alpha = .90$).

Anthropomorphism

The Godspeed questionnaire (Bartneck et al., 2008, German version was obtained from 2019) was used to examine anthropomorphism, the attribution of human qualities to non-human entities. The 5-point semantic differential scale has four items, asking participants to rate the profiles as either machinelike versus humanlike, artificial versus lifelike, fake versus natural, and conscious versus unconscious. The scale showed excellent internal consistency ($\alpha = .93$).

Rating of Responses (Appeal/Liking)

Participants rated the appeal of each of the match’s 10 ICGT responses immediately after reading it. For each response, they indicated “How appealing do you find this answer?” on a 7-point Likert scale ranging from 1 (not appealing at all) to 7 (very appealing). To operationalize overall liking of the match’s answers, ratings were averaged across the 10 responses, with higher scores indicating greater perceived appeal of the response set.

Demographics

Age was assessed with an open-ended question, while gender and sex were assessed using both open and multiple-choice items.

**Note that additional measures were used, but not included in the following analysis (e.g., mate value and Mini-IPIP).*

Manipulation Check

A manipulation check was conducted to ensure that the vignette was successful and that the source label conditions were perceived to be different from each other. The manipulation check showed significant differences between the AI and human conditions on the four anthropomorphism items of Bartneck's Godspeed questionnaire (Bartneck et al., 2019). People who were matched with an AI partner rated the answers as less natural ($t(305) = -3.06, p = .002$), humanlike ($t(305) = -3.29, p < .001$), conscious ($t(305) = -4.81, p < .001$), and lifelike ($t(305) = -2.74, p = .007$) than those who were matched with a human. Those participants rated the answers as more natural, humanlike, conscious, and more lifelike (see Table 1 for descriptives).

Results

To avoid overfitting and unnecessary model complexity, potential covariates were evaluated prior to the main analyses. Following recommendations to include covariates only when they show a meaningful association with the dependent variables (Miller & Chapman, 2001), two multiple linear regression analyses were conducted to assess whether loneliness, attitudes toward AI, liking of the response content, and age predicted interpersonal closeness and interest in intimate interactions. Only variables demonstrating a significant contribution were retained as covariates in subsequent analyses. Both regression models were significant (closeness: $R^2 = .518, F(4, 302) = 80.98, p < .001$; intimate interest: $R^2 = .467, F(4, 302) = 66.23, p < .001$; see Tables 2 and 3). In both models, only rating of the responses (liking/appeal) emerged as a significant predictor (closeness: $\beta = .72, p < .001$; intimate interest: $\beta = .68, p < .001$), whereas age, loneliness, and attitudes toward AI were not associated with the outcomes. Therefore, rating of the responses was retained as the sole covariate.

A factorial MANCOVA with source label, gender, and nationality as between-subject factors and liking of the responses as a covariate was conducted to examine effects on interpersonal closeness and intimate interest. In line with the hypotheses, the model included the two-way interactions of ontological class (human vs. AI) with gender and nationality, as well as the three-way interaction (ontological class $\times$ gender $\times$ nationality). The assumption of covariance homogeneity was met (Box's $M = 1.359, F = 0.450, p = .717$). Similarly, the assumption of variance homogeneity was met, as Levene's Test for Equality of Variances was nonsignificant for both dependent variables (intimate interest: $F(7, 299) = 1.612, p = .132$; closeness: $F(7, 299) = 0.525, p = .816$).

There was no significant main effect of source label on the combined dependent variables ($F(2, 297) = 0.070, p = .933, \eta^2 < .000$). Consistent with this, participants in the human-label condition did not report higher interpersonal closeness than those in the AI-label condition ($F(1, 302) = 0.118, p = .731, \eta^2 < .000$; estimated marginal means: AI label $M = 3.942, SE = 0.084, 95\% CI [3.777, 4.106]$; human label $M = 3.986, SE = 0.095, 95\% CI [3.798, 4.173]$) and did not report higher interest in intimate interactions ($F(1, 302) = 0.032, p = .858, \eta^2 < .000$; estimated marginal means: AI label $M = 3.062, SE = 0.058, 95\% CI [2.947, 3.177]$; human label $M = 3.078, SE = 0.067, 95\% CI [2.947, 3.209]$). Thus, the hypothesis that human (vs. AI) source labeling increases closeness and intimate interest was not supported.

TABLE 1
Descriptive Statistics of Main Variables

Scale	Cultural Dimensions	Entity	<i>M</i>	<i>SD</i>	<i>N</i>
Inclusion of the Other in the Self	General	Human	4.11	1.55	136
		AI	3.85	1.63	171
Interest in Intimate Interactions	General	Human	3.17	1.00	136
		AI	3.00	1.08	171
Liking of the Answers	General	Human	5.39	1.04	136
		AI	5.09	1.20	171
	Germany	Human	5.18	1.00	62
		AI	4.78	1.29	79
	Canada	Human	5.56	1.05	74
		AI	5.35	1.05	92
Anthropomorphism: All Dimensions	General	Human	3.27	1.19	136
		AI	2.76	1.34	171
Anthropomorphism: Fake/Natural	General	Human	3.10	1.32	136
		AI	2.65	1.25	171
Anthropomorphism: Machinelike/Humanlike	General	Human	3.31	1.39	136
		AI	2.82	1.22	171
Anthropomorphism: Conscious/Unconscious	General	Human	3.61	1.16	136
		AI	2.96	1.19	171
Anthropomorphism: Artificial/Lifelike	General	Human	3.16	1.41	136
		AI	2.73	1.33	171
Loneliness	General	Human	2.46	0.93	136
		AI	2.60	0.92	171
Attitudes Toward AI	General	Human	3.30	0.68	136
		AI	3.32	0.62	171

TABLE 2
Regression Analysis: Age, Loneliness, and Liking of the Answers
Predicting Interpersonal Closeness

Predictor	<i>B</i>	<i>SE</i>	β	<i>t</i>	<i>p</i>	95% CI		VIF
						<i>LL</i>	<i>UL</i>	
Constant	−1.101	.553	–	−1.993	.047	−2.189	−.013	–
Age	.004	.006	.024	.596	.552	−.008	.016	1.056
Loneliness	−.022	.069	−.013	−.319	.750	−.157	.113	1.009
Liking of Answers	.995	.055	.718	17.96	< .001	.887	1.103	1.001

Not. CI = confidence interval; LL = lower limit; UL = upper limit; VIF = variance inflation factor. * $p < .001$

TABLE 3
Exploratory Multiple Regression Predicting Liking of the Answers (AI Label Condition)

Predictor	<i>B</i>	<i>SE</i>	β	<i>t</i>	<i>p</i>
Constant	3.024	0.475	—	6.369	< .001
Fake/Natural	0.479	0.112	0.503	4.275	< .001
Machinelike/Humanlike	−0.079	0.116	−0.081	−0.684	.495
Unconscious/Conscious	0.466	0.098	0.462	4.772	< .001
Artificial/Lifelike	−0.245	0.110	−0.274	−2.228	.027
Loneliness	0.099	0.079	0.076	1.247	.214
Attitudes Toward AI	0.013	0.119	0.007	0.109	.914

Note. Condition: Ontological Class = AI. Dependent Variable: Liking of Answers.

There was also no significant interaction effect between source label and participants' gender ($F(2, 297) = 1.076, p = .342, \eta^2 = .007$) or nationality ($F(2, 297) = 0.117, p = .890, \eta^2 < .001$), and the three-way interaction was likewise nonsignificant ($F(2, 297) = 1.937, p = .146, \eta^2 = .013$). However, the MANCOVA revealed a strong effect of the covariate liking of the responses on the combined dependent variables (Wilks' $\Lambda = .433, F(2, 297) = 194.288, p < .001, \eta^2 = .567$).

Exploratory Analyses

A linear regression analysis was conducted to identify the key factors influencing participants' liking of the AI match's answers, as this insight informs our understanding of what drives positive perceptions of AI-generated responses. Loneliness, general attitudes toward AI, and the four items of the anthropomorphism questionnaire were put in as predictors, $R^2 = .390, F(6, 163) = 17.343, p < .001$ (see Appendix, Table 4 and Table 5). Three of the four anthropomorphism items significantly predicted the rating of the answers of the AI match, namely how natural versus fake ($p < .001$), conscious versus unconscious ($p < .001$), and how lifelike versus artificial ($p = .027$) the AI match was perceived by the participants. Hence, perceiving the AI match as more natural, conscious, and lifelike increased liking of its answers.

Discussion

This study examined whether people could form interpersonal closeness and intimate interest toward a potential partner labeled either as human or AI based on AI-generated, self-disclosure messages within a mock AI matchmaking scenario across gender (women and men) and nationality (German and Canadian). Results showed no significant difference in closeness or interest based on the source label, gender, or nationality. Instead, liking the AI-generated text, which was associated with perceived human-likeness, was the key factor. The findings point to the relevance of communication qualities, rather than perceived or labeled agent identity, in shaping early impressions of interpersonal closeness.

TABLE 4
Exploratory Multiple Regression Predicting Liking of the Answers
(Human Label Condition)

Predictor	<i>B</i>	<i>SE</i>	β	<i>t</i>	<i>p</i>
Constant	5.007	0.511	—	9.807	< .001
Fake/Natural	0.092	0.142	0.116	0.649	.517
Machinelike/Humanlike	0.073	0.121	0.097	0.608	.544
Unconscious/Conscious	0.271	0.118	0.300	2.288	.024
Artificial/Lifelike	−0.081	0.150	−0.108	−0.540	.590
Loneliness	0.044	0.093	0.039	0.471	.638
Attitude Toward AI	−0.296	0.126	−0.194	−2.354	.020

Note. Condition: Ontological Class = Human. Dependent Variable: Liking of the Answers.

TABLE 5
Regression Analysis: Age, Loneliness and Liking of the Answers and Attitude
Toward AI Predicting Interest in Interpersonal Interaction

Predictor	<i>B</i>	<i>SE</i>	β	<i>t</i>	<i>p</i>	95% CI		VIF
						LL	UL	
Constant	−1.015	1.54	—	−.659	.510	−4.046	2.016	—
Age	−0.002	.017	−.006	−.132	.895	−.035	.032	1.056
Loneliness	.171	.192	.038	.892	.374	−.206	.548	1.009
Liking of Answers	2.504	.154	.681	16.213	< .001	2.200	2.808	1.001
Attitude toward AI	−.025	.280	−.004	−.088	.930	−.577	.527	1.052

Note. CI = Confidence Interval; LL = lower limit; UL = upper limit; VIF = variance inflation factor.

Interpretations

Source Label

The lack of effect of the source label on interpersonal closeness and intimate interest can be interpreted in several ways. For one, it may suggest that people rely less on the perceived, source labeled ontology of an agent than on their appreciation of its produced output (e.g., liking the content of the communication) when bonding or considering intimacy with a potential partner. These findings align with prior work indicating that evaluations in HMC can be strongly shaped by message-level qualities and interaction cues rather than categorical source labels (A. Ho et al., 2018). More broadly, they resonate with HMC's conceptual emphasis on communication as a trans-ontological, interpretive, and relational accomplishment rather than a function of fixed ontological categories, as Edwards and Etzrodt (2025) conclude "HMC's significance lies in its capacity to illuminate and engage two interrelated dynamics: the act of communicating across ontological boundaries and

the interpretive labor involved in making such communication intelligible and meaningful” (p. 8). This framing is important for the present work because it focuses precisely on early meaning-making under ontological ambiguity: participants encounter identical self-disclosure messages under different ontological labels and must render the communicator intelligible and relationally meaningful based on limited, text-based cues. They further align with research on intimacy with technology suggesting that interest in, and the development of, intimate human-machine relationships may depend more on the quality of the experience (e.g., positive or negative communication) than on the nature of the agent itself (Dubé & Anctil, 2020). These findings are also in line with work questioning the primacy of interhuman intimacy, highlighting that bonds and intimate exploration can emerge with potential partners as long as a certain standard of (positive) features and experience is met (Dubé & Anctil, 2020).

Extending this literature, the fact that both conditions elicited interpersonal closeness and intimate interest in some participants suggests that, in text-based contexts where machine-generated communication can be experienced as highly human-like, relational impressions may be shaped less by categorical labels and more by how satisfying the initial exchange is. This interpretation aligns with theorizing in HMC that emphasizes trans-ontological communication and the interpretive work through which communicative partners become intelligible and relationally meaningful (Edwards & Etzrodt, 2025), as well as research on intimacy with technology highlighting the role of interaction quality in motivating continued engagement and relational development (Dubé & Anctil, 2020). Such satisfaction appears related to human-likeness (i.e., perceived naturalness, consciousness, and lifelikeness). This aligns with Pentina and colleagues (2023) who found that human-likeness and authenticity (i.e., agency, autonomy, and uniqueness) contribute to chatbot acceptance. Consistent with this, first qualitative interviews with individuals who report ongoing chatbot partners found that users actively use romantic fantasy to shape interactions and make the chatbot feel “as human as possible.” Accompanying survey results found romantic fantasy to be the strongest predictor of relationship intensity, which could help alleviate the technological limitations of the chatbot partner and make the relationship seem more encompassing (Ebner & Szczuka, 2025).

Notably, although several studies have reported differences in evaluations of, and reactions to, intimate communication with humans versus artificial entities, the technologies and contexts employed in prior work differ from this study. For instance, many earlier studies used more sexualized or flirtatious contexts and relied on visual stimuli depicting artificial partners (cf., Banks & Van Ouytsel, 2020; Jiang et al., 2013; Szczuka et al., 2019). Text-generative AI, however, may operate differently. As a hybrid technology (i.e., artificial outputs trained on human data), LLMs may partly reflect human responses convincingly enough to approximate human communication in text-based scenarios. In line with van der Goot and Etzrodt’s (2023) conceptualization of technological entities as existing in a liminal space, described as “betwixt and between,” the blurring of human AI boundaries may shift attention away from the labeled ontology of the communicator toward the qualities of the communication itself.

This shift is particularly likely in text-dominant tasks, where judgments of interpersonal closeness and intimate interest must be formed in the absence of visual, behavioral, or interactional cues. Under these conditions, both human and AI matches may elicit similar

levels of interpersonal closeness and intimate interest, not necessarily because they are evaluated as equally appealing, but because the limited information available constrains differentiation. As a result, the human partner may appear less appealing than anticipated, the AI partner more appealing than expected, or both. After all, engaging intimately with humans may involve risks (e.g., physical or emotional harm), whereas artificial partners may offer advantages, such as continuous judgment-free interactions and reduced risks of harm (e.g., injuries or infections; Dubé & Anctil, 2020). Some people may also be curious about artificial partners, and such technologies may offer novelty (Croes & Antheunis, 2021).

It is also possible that neither allegedly perfect match (AI and human) was perceived as particularly compelling, leading participants to respond cautiously in both conditions. Developing interpersonal closeness and deciding to pursue intimate engagement typically requires richer and more differentiated information than brief self-disclosure prompts, such as those drawn from the ICGT. Moreover, LLMs are designed to produce statistically likely and socially appropriate responses, which may result in profiles that appear coherent and pleasant, yet also relatively generic. This tendency toward stereotypical or normative answers may reduce distinctiveness, making neither the human nor the AI partner stand out as especially unique or memorable.

Gender and Nationality

Given the consistent gender effects found in previous research on intimacy with technology, the lack of gender differences in this study was surprising (cf., Bame, 2017; Brandon et al., 2022; Brandon & Planke, 2021; Dubé et al., 2023, 2024; Hanson, 2013; Nguyen, 2020; Nordmo et al., 2020; Oleksy & Wnuk, 2021; Scheutz & Arnold, 2016). It may suggest that in certain contexts, such as a text-based matchmaking scenario, women and men behave more similarly than expected. This may be due to several reasons.

For one, previous empirical studies typically used still images or hypothetical written scenarios in which participants had to imagine artificial partners (e.g., sex robots) or interactions with them (e.g., Nordmo et al., 2020; Scheutz & Arnold, 2016). These studies offered limited to no information about the artificial partner's profile or personality, leading participants to rely on the physical cues of the stimuli. And since women tend to rely more on contextual cues and rate visual stimuli as less arousing than men, this limited information may have generated a gender difference (Rupp & Wallen, 2008). Participants may also have relied on media depictions that overwhelmingly feature female artificial partners and male-gynoid relationships, as well as social norms that favor men's intimacy and sexuality and shame women's sexual agency (Dubé et al., 2024).

In contrast, the self-disclosure in this study provided participants with additional information about their potential partners, offering some insights into their personality and individual characteristics. Given the substantial overlap in what women and men seek in intimate partners (e.g., kindness, commitment, intelligence; Buss, 2016), this information may have narrowed gender differences in connection and interest toward human-machine intimacy. Moreover, because intimate and sexual relationships often entail unbalanced, gendered risks (e.g., physical harm, unwanted pregnancy), women must typically rely on additional information (e.g., personality indicators) to evaluate safety and willingness to engage intimately with a partner (Buss, 2016): information that was more available here than in previous studies. Some evidence also suggests that men rely more on visual cues,

whereas women rely more on contextual and background information when in intimacy-related decisions and reactions (Buss, 2016; Rupp & Wallen, 2008), partly explaining gender differences in pornography versus erotic literature consumption (Solano et al., 2020), as well as accounting for the prior and current study findings. Finally, it is possible that gender plays limited to no role in some contexts, including such highly text-based situations.

Similarly, the lack of nationality effects may reflect the limited role of this variable in the current context. However, given previous research on cultural differences in attitudes toward machines and artificial partners (Folk et al., 2025), it is perhaps more likely that, as part of two Western cultures, Germans and Canadians are more similar than initially hypothesized with regard to interpersonal closeness and intimate interest with machines. Indeed, beyond indulgence-restraint and uncertainty avoidance, Hofstede's cultural dimensions (Hofstede & Bond, 1984) also indicate that Canada and Germany are relatively close on other domains, such as individualism, both cultures emphasize personal autonomy, self-expression, and self-directed decision-making, as well as long-term orientation, which shapes future-oriented thinking and may influence openness to technological developments. These shared characteristics may have fostered comparable comfort levels and willingness to consider relational or intimate connections with AI partners, resulting in the observed lack of effect.

Theoretical and Methodological Implications

The present findings resonate with theories in HMC. They echo key tenets of early HMC paradigms, such as Media Equation Theory and Computers as Social Actors (Nass & Moon, 2000; Reeves & Nass, 1996) which propose that people may respond similarly to machines and humans. However, more consistent with newer paradigms, such as Media Evocation and AMR, these findings point to a reflective loop in which people not only consider the communicative partner's ontology, but also interpret its actions, their meaning, and the relational and contextual engagement they entail (Edwards & Etzrodt, 2025). In light of such paradigms, the present findings suggest that, in an online matchmaking scenario (context) involving asynchronous text-based self-disclosure (action), interpersonal closeness and willingness to engage intimately with an AI partner appear primarily linked to how much participants liked the communication. This evaluation was itself strongly associated with perceived anthropomorphism of the agent, or how natural, conscious, and lifelike it is considered (meaning). This indicates that people do not automatically or mindlessly respond to machines as they respond to humans. Instead, they engage in reflective, interpretive processes, even when their apparent behaviors are the same.

These findings also resonate with theories on intimacy with technology. Aligned with the HEICEM, they underscore the primacy of interactive qualities and meaning attribution in human-machine erotic interactions and co-evolution (Dubé & Ancil, 2020). As predicted by the HEICEM, they suggest that more positive interactions (e.g., enjoyable communications) will increase the likelihood of future interactions and the development of human-machine intimacy (Dubé & Ancil, 2020).

Research on intimacy-oriented HMC benefits from explicitly aligning design choices with the relational phenomena under study. In one-off, asynchronous, text-based paradigms, judgments of interpersonal closeness and intimate interest may be shaped strongly

by message-level cues and vignette characteristics, potentially attenuating or obscuring more subtle effects of categorical source labels (A. Ho et al., 2018). At the same time, these paradigms can be useful for isolating early meaning-making under ontological ambiguity: they tightly control exposure and examine how participants render a communicator intelligible and relationally meaningful from minimal, text-only cues. By design, they bracket longer-term, interaction-dependent processes (e.g., reciprocity building across repeated exchanges) rather than attempting to approximate them within a single exposure. This emphasis mirrors broader HMC conversations that foreground questions of methodological validity and the limits of demonstrational designs in a rapidly evolving technological landscape (Etzrodt et al., 2024).

Although the manipulation successfully elicited differences in perceived anthropomorphism, we cannot conclusively determine how participants made sense of the communicator's ontological status beyond the provided label. This reflects HMC's view that ontological classifications are communicatively constructed and can be shaped by relational performance rather than by fixed attributes alone (Edwards & Etzrodt, 2025). However, because perceived human-likeness was not experimentally manipulated but measured, associations between anthropomorphism, liking of the responses, and relational outcomes should be interpreted as correlational rather than causal. Accordingly, the present findings do not allow conclusions about directional effects of anthropomorphism but rather indicate that perceived human-likeness co-occurs with more positive evaluations of communication and relational impressions in this context.

Social and Practical Implications

While the present experiment can speak to early relational impressions in a controlled, asynchronous, text-based context, broader societal implications necessarily require interpretation across levels of analysis. We therefore articulate the following points as plausible, evidence-informed implications that motivate focused future research and design-governance measures. The findings of this study have important implications for online dating and human-machine intimate relationships. They suggest that individuals can develop situational feelings of interpersonal closeness and intimate interest toward AI-generated dating profiles, provided that the communicative content is perceived as appealing and human-like. This aligns with prior work showing that AI-mediated dating interactions can elicit meaningful relational impressions even in the absence of sustained reciprocity or long-term interaction (J. Q. Ho et al., 2025; Pentina et al., 2023).

At the same time, these findings raise concerns regarding the potential misuse of generative AI in intimate communication contexts. Results from the manipulation check indicate that merely labeling identical messages as human is sufficient to increase perceived human-likeness, suggesting that source labels alone can meaningfully shape first impressions even when communicative content is fully machine-generated. In combination with increasingly capable LLMs that produce statistically optimized, socially fluent, and context-sensitive text, this creates opportunities for impersonation, strategic self-presentation, and manipulation in online dating environments (Bartaula, 2025; J. Q. Ho et al., 2025). Such practices may range from relatively benign outsourcing of message formulation to more problematic uses, including the fabrication of personas, catfishing, or other

forms of deceptive relational engagement. Prior research has highlighted that AI-powered dating infrastructures do not merely facilitate interaction but actively shape how intimacy, attraction, and relational decision-making are structured and evaluated, often in ways that remain opaque to users (Wang, 2023). From this perspective, risks such as romance-related scams and other forms of intimate cybercrime are not peripheral issues but structural vulnerabilities emerging from AI-mediated dating ecosystems (Bartaula, 2025).

Beyond relational harms, the use of AI-generated content in dating contexts also introduces concrete cybersecurity risks, including social engineering, identity fraud, and the scalable exploitation of emotional vulnerability. Because AI-assisted profiles and messages can be rapidly generated, adapted, and personalized, such systems may lower the threshold for large-scale deceptive practices, underscoring the need to treat intimate communication environments as security-relevant infrastructures rather than purely social spaces. Over time, widespread reliance on AI-assisted intimate communication may erode trust in online dating and interpersonal communication more broadly, as users become increasingly uncertain about the source, effort, and authenticity of relational messages. This uncertainty may arise when individuals partially or fully outsource self-disclosure to AI systems, embellish personal information, or interact with fabricated profiles designed to exploit emotional vulnerability (J. Q. Ho et al., 2025). In parallel, extensive reliance on AI for intimate communication may shape how relational and emotional skills are practiced, as aspects of interpersonal interaction are increasingly externalized to technology, potentially altering comfort, confidence, and engagement in human–human relationships (Wang, 2023).

These dynamics underscore the need to integrate principles of responsible AI use into the design and governance of online dating services. Given that online dating has become one of the most prevalent infrastructures for forming intimate and long-term relationships, the deployment of AI in these contexts warrants particular ethical and cybersecurity scrutiny (Wang, 2023). Primary responsibility lies with platform providers and technology companies, who design, deploy, and profit from AI-mediated communication systems and therefore bear an obligation to ensure transparency, fairness, and user protection by design. This includes addressing risks related to deception, impersonation, and the exploitation of AI-mediated intimacy for malicious purposes (Bartaula, 2025). At the same time, and precisely because intimate communication increasingly involves user-generated and AI-assisted content, complementary efforts are needed to raise user awareness that interactions may be partially or fully mediated, optimized, or generated by AI. Prior research suggests that such awareness is crucial for mitigating misattributions of authenticity and intentionality in HMC (J. Q. Ho et al., 2025). Protecting users in AI-enhanced dating environments therefore requires a dual approach that combines strong platform-level accountability with measures that enable informed, reflective, and security-conscious use by individuals.

Finally, beyond online dating, the present findings contribute to broader discussions of human-machine intimacy. They suggest that text-based machine communication has reached a level at which it can rival human communication in shaping early relational impressions, and that many individuals are open to forming or exploring intimate connections with artificial partners when the communicative output resonates with them. This may help explain the growing popularity of AI companions and foreshadows an expanded

role of artificial agents in intimate life, particularly as their interactive capacities extend beyond language and as the meanings attributed to these agents and relationships continue to evolve (Dubé & Anctil, 2020).

Limitations and Strengths

This study presents limits. For one, it used an adapted version of the ICGT (Aron et al., 1997) to fit the constraints of an online format. The task was shortened and conducted asynchronously, lacking the real-time interaction of the original measure. These adjustments align with asynchronous online communications (e.g., social media or dating platforms), but they do not replicate the in-person dynamics of interpersonal closeness and intimate interest and limit our ability to generalize the findings to them. This study was also limited in gender, sexual orientation, language, countries, and sociocultural backgrounds, which constrains generalizability to larger populations. It also took place at a single time point, limiting our ability to assess the effects of repeated or long-term exposure and interactions with AI partners.

Finally, and particularly relevant to the present research questions, a notable number of participants had to be excluded due to errors in the manipulation check. This suggests that some participants may have missed, misunderstood, or discounted the source label manipulation (i.e., whether the profile was labeled as human or AI). This raises the possibility that the source label did not function as a salient ontological cue for a substantial subset of participants. Prior research suggests that categorical labels such as “chatbot” or “AI” do not necessarily determine users’ relational and emotional responses. For instance, A. Ho et al. (2018) found that emotional and relational outcomes of disclosure did not differ depending on whether participants believed they were interacting with a chatbot or a human partner. Importantly, this limitation can also be interpreted in line with the present findings. Rather than solely indicating a methodological weakness, the limited salience of the source label manipulation may reflect a broader boundary condition: in asynchronous, text-only encounters, categorical source labels may be insufficient to meaningfully shape perceptions of interpersonal closeness, liking, or intimate interest. In such contexts, participants may prioritize what the message does communicatively over what the source is labeled to be, consistent with relational and communicative perspectives on human-machine interaction.

Still, this study advances research in HMC and intimacy with and through technology. It is the first experimental study examining whether people can develop interpersonal closeness and intimate interest toward human and AI partners in a mock matchmaking scenario where all communications were AI-generated. Its findings suggest that people across gender (i.e., men and women) and nationality (German and Canadian) can form connections with both humans and machines. In that regard, liking the communications, likely based on the perceived anthropomorphism of the agent, seems to be more important than who the communicator is. Moreover, this study’s findings hint that, in certain contexts, where information about a potential partner is more obvious, typical gender differences in attitudes toward human-machine intimacy seem to disappear. This calls for further consideration of the stimuli and context used in research on HMC and human-

machine intimate interaction. Similarly, the potential boundary condition in categorical label effects invites HMC research to further (re)consider how and when to use source labels in manipulation, and what these labels actually manipulate.

Future Directions

Future research in this area may want to employ more interactive methods, such as live videos, VR tasks, or emerging more sophisticated chatbots, to replicate the results and improve the experimental design of this study. Such studies could also test interactions over a longer period to assess long-term effects of HMC in intimate contexts with LLM-powered agents. To better capture people's interpretation of AI partners and why they liked the content or interactions (or not), future studies may want to add qualitative components, such as interviews and open-ended questions. These studies should also aim to include more diverse populations, such as individuals from LGBTQIA+ communities and other national, geographical, and cultural contexts to improve finding generalizability. This research may further examine more culturally distinct countries than Canada and Germany (e.g., Japan and the United States) and focus on individual sociocultural backgrounds in order to assess how they shape people's responses to potential human and AI partners, including in interpersonal and intimate contexts. This research is necessary as AI becomes more prevalent in our intimate communications and relationships.

Conclusion

This study demonstrates that interpersonal closeness and intimate interest can emerge from LLM generated self-disclosure, largely independent of whether a potential partner is labeled as human or AI. Rather than source labels, participants' reactions were primarily associated with the perceived quality and human-likeness of the responses, underscoring text-based communication as a powerful site for the construction of social meaning. These findings challenge conventional assumptions about authenticity in online dating and show how relational impressions can be increasingly shaped by AI-mediated communication. At the same time, they carry clear governance implications. As online dating represents one of the most central infrastructures for forming intimate relationships and AI partners become more prevalent in our intimate lives, the capacity of LLMs to evoke closeness and intimacy calls for responsible AI use by design.

CRedit Author Contribution Statement

Jessica M. Szczuka: Conceptualization, Methodology, Supervision, Formal analysis, Writing original draft, Writing review and editing.

Lisa Mühl: Methodology, Data collection, Data curation, Writing review and editing.

Paula Ebner: Data collection, Data curation, Formal analysis, Writing review and editing.

Simon Dubé: Conceptualization, Methodology, Supervision, Writing review and editing.

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